

BARNWELL AGDM, AB, Canada

WMO: 713460

Lat: 49.8007N

Lon: 112.3026W

Elev: 824

StdP: 91.81

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.5	-25.3	-32.3	0.2	-28.5	-28.9	0.3	-25.2	15.9	6.2	14.1	4.1	2.7	230	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	31.8	17.3	29.8	16.7	28.0	16.3	19.3	27.8	18.2	26.8	17.2	25.7	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	12.8	23.2	15.1	11.8	21.7	13.9	10.9	20.5	58.7	27.8	54.9	27.0	51.5	25.7	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 14.3	12.5	10.7	DB	-33.6	35.1	3.1	2.1	-35.8	36.6	-37.7	37.9	-39.4	39.0	-41.6	40.6	(4)
(5)			WB	-33.8	22.2	3.0	1.4	-35.9	23.2	-37.6	24.1	-39.3	24.9	-41.4	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-5.6	-5.9	-0.6	5.9	11.2	15.4	18.8	17.7	13.0	-0.6	-6.0	(6)	
(7)		DBStd	10.99	8.90	8.68	8.29	4.98	4.54	3.13	2.83	3.25	4.50	5.47	7.63	8.53	(7)
(8)		HDD10.0	2413	484	447	331	137	40	1	0	1	21	133	321	497	(8)
(9)		HDD18.3	4643	742	680	588	374	222	98	28	51	167	369	568	756	(9)
(10)		CDD10.0	904	0	0	2	13	78	163	274	239	110	23	2	0	(10)
(11)		CDD18.3	92	0	0	0	0	2	9	43	31	6	0	0	0	(11)
(12)		CDH23.3	2022	0	0	1	12	114	211	789	658	224	11	0	0	(12)
(13)		CDH26.7	685	0	0	0	2	27	49	286	246	76	1	0	0	(13)
(14)	Wind	WSAvg	4.7	5.3	4.6	5.0	5.2	4.9	4.5	3.8	3.7	4.3	4.8	5.2	5.0	(14)
(15)	Precipitation	PrecAvg	369	13	10	18	29	53	90	39	35	32	21	15	13	(15)
(16)		PrecMax	514	27	22	60	61	105	227	117	78	110	67	35	27	(16)
(17)		PrecMin	255	1	0	0	8	16	14	2	1	3	4	2	0	(17)
(18)		PrecStd	68	7	6	12	13	24	50	29	21	26	14	9	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	13.7	18.6	25.2	29.6	31.0	34.2	33.7	32.2	25.0	18.0	12.0	(19)
(20)			MCWB	6.9	5.8	8.6	11.5	14.8	18.2	18.2	17.1	15.5	12.3	9.5	5.4	(20)
(21)		2%	DB	9.7	9.8	15.5	21.2	26.6	27.8	31.9	31.7	29.0	21.3	13.8	9.2	(21)
(22)			MCWB	4.8	4.1	7.3	9.9	13.7	16.4	18.7	16.4	14.7	11.2	7.1	4.4	(22)
(23)		5%	DB	7.5	7.4	13.0	18.2	23.8	25.8	29.8	29.5	25.8	18.1	11.4	7.0	(23)
(24)			MCWB	3.3	2.7	5.7	8.3	12.4	15.6	18.0	16.1	13.6	9.8	5.9	2.8	(24)
(25)		10%	DB	5.3	5.1	10.4	15.4	21.1	23.6	27.8	27.2	22.6	15.3	9.2	5.1	(25)
(26)			MCWB	2.1	1.4	4.5	7.0	11.2	14.3	17.4	15.7	12.6	8.2	4.3	1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	6.3	9.4	12.0	15.6	19.4	22.2	19.2	16.9	13.4	10.1	6.2	(27)
(28)			MCDWB	11.8	11.7	17.3	23.0	27.1	27.7	30.0	28.5	28.0	22.8	17.3	10.7	(28)
(29)		2%	WB	4.9	4.4	7.8	10.4	14.0	17.7	20.2	18.0	15.3	11.7	7.7	4.6	(29)
(30)			MCDWB	9.4	9.5	14.5	20.2	24.8	25.7	28.6	27.6	26.4	20.0	13.1	8.7	(30)
(31)		5%	WB	3.5	2.9	6.2	8.8	12.8	16.3	19.0	17.2	14.2	10.1	6.2	3.0	(31)
(32)			MCDWB	7.1	7.1	12.4	16.9	22.3	23.8	27.3	26.7	24.6	17.3	11.1	6.5	(32)
(33)		10%	WB	2.2	1.5	4.7	7.5	11.7	15.1	18.1	16.2	13.0	8.7	4.6	1.6	(33)
(34)			MCDWB	5.2	4.9	10.0	14.6	19.7	22.1	26.3	25.4	21.7	14.7	8.8	4.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.0	11.9	12.6	13.5	13.8	13.3	16.0	16.2	15.0	13.4	11.7	10.8	(35)
(36)			MCDBR	11.5	13.9	16.1	19.5	18.9	17.5	19.9	21.4	21.1	17.8	13.9	11.4	(36)
(37)			MCWBR	7.8	9.0	9.1	9.7	8.2	7.9	8.8	8.4	8.9	9.1	8.4	7.6	(37)
(38)		5% WB	MCDWB	10.9	13.5	15.6	17.6	17.1	15.9	17.8	18.2	19.9	17.1	13.7	11.7	(38)
(39)			MCWBR	7.6	9.2	9.3	9.1	7.7	7.8	8.8	7.9	8.8	8.9	8.7	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.268	0.290	0.327	0.341	0.339	0.342	0.359	0.319	0.289	0.262	0.245	(40)	
(41)		taud	2.454	2.482	2.473	2.375	2.383	2.437	2.443	2.370	2.456	2.519	2.497	2.445	(41)	
(42)		Ebn at Noon	816	898	929	920	916	918	910	875	882	857	806	782	(42)	
(43)		Edh at Noon	59	73	89	110	114	109	107	109	90	70	56	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.24	2.24	3.52	4.86	5.78	6.41	7.02	5.60	3.98	2.47	1.38	0.97	(44)	
(45)		RadStd	0.10	0.18	0.28	0.33	0.46	0.44	0.36	0.35	0.34	0.25	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page